Sophia Sun

1 (617) 777-0906 | shs066@ucsd.edu | huiwenn.github.io

Education

University of California San Diego, San Diego, CA — Ph.D. 2018 - 2025
Thesis: Probabilistic Deep Learning for Uncertainty Quantification and Decision Making
Artificial Intelligence Group, Computer Science and Engineering
Advisor: Prof. Rose Yu

Wellesley College, Wellesley, MA — B.A. 2014 – 2018
Double Major in Mathematics and Computer Science.

Massachusetts Institute of Technology, Cambridge, MA — *Cross Registration* 2016 – 2018
Coursework in Mathematics and Computer Science

Experiences

Applied Scientist, AWS AI Fundamental Research Santa Clara, CA 2025 – present

(Agentic Systems, RAG, Knowledge and Context Layer, Multi-agent harness and evaluation)

* Built personalized knowledge graph and context layer for amazon Quick AI agents, improving Quick's multi-hop question-answering accuracy by 25% on internal benchmarks.
* Built amazon Quick's end-to-end evaluation harness, bringing insight into multiple layers of the agentic system: cost and functionality, agent behavior, task goal success, user satisfaction. Authored benchmarks that are run every week /every release to identify regressions and improvements.

Applied Scientist Intern, AWS AI Santa Clara, CA 2023, 2024

(time-series forecasting, uncertainty quantification, anomaly detection)

* Designed unsupervised **anomaly detection** algorithm for AWS RDS usage through conformal prediction and quantile tracking methods, eliminating errors in the cold start period of the RDS anomaly detection pipeline. paper

(ambiguity detection, memory in LLM agents)

* Designed and implemented pipeline to handle **ambiguities in the LLM**-powered AWS Text-to-SQL product. paper

Research Scientist Intern, Amazon Robotics RAD 2020

(reinforcement learning, multi-agent planning, optimization)

* Machine learning for prediction-based multi-robot trajectory planning in amazon's fulfillment centers.
* Help built simulation for multi-robot trajectory planning and improved overall throughput by 3% in simulation.

Intern, Google DeepMind London, U.K. 2018

(backend software engineering, data science)

* Secure Electronic Health Records data pipelining, analysis, and verification.
* Improved interpretability and verifiability of analytical algorithms deployed in Deepmind's Streams App.

Publications

Machine Learning and Uncertainty Quantification

Sophia Sun. "Probabilistic Deep Learning for Uncertainty Quantification and Decision-Making." *Doctoral dissertation, University of California, San Diego.* 2025

Sophia Sun, Rose Yu. "Conformal Prediction for Time-series Forecasting with Change Points." In proceedings of *Neural Information Processing systems. NeurIPS 2025*

Sophia Sun, Abishek Sankararaman, Murali Narayanaswamy. "Online Adaptive Anomaly Thresholding with Confidence Sequences." In proceedings of *International Conference on Machine Learning, ICML* 2024

Sophia Sun, Rose Yu. "Copula Conformal Prediction for Multi-step Time Series Forecasting." In proceedings of *International Conference on Learning Representations ICLR*, 2024

Sophia Sun, Wenyan Chen, Zihao Zhou, Sonia Fereidooni, Elise Jortberg, Rose Yu. "Data-Driven Simulator for Mechanical Circulatory Support with Domain Adversarial Neural Process." In proceedings of *Learning for Dynamics and Control Conference (L4DC),* 2024

Sophia Sun, Robin Walters, Jinxi Li, Rose Yu. "Probabilistic Symmetry for Multi-Agent Dynamics." In proceedings of *Learning for Dynamics and Control Conference (L4DC),* 2023

Large Language Models

Sophia Sun, Abishek Sankararaman, Murali Narayanaswamy. "Resolving Ambiguity through Personalization in LLM chat systems." Workshop on Reasoning and Planning for Large Language Models @ *ICML* 2025

Bodhisattwa Prasad Majumder, Shuyang Li, Jianmo Ni, Huanru Henry Mao, **Sophia Sun**, Julian McAuley. "Bernard: A Stateful Neural Open-domain Socialbot." *Alexa Prize Proceedings*, 2020

Reinforcement Learning & Robotics

Aysin Tumay*, **Sophia Sun***, Sonia Fereidooni, Aaron Dumas, Rose Yu. "Uncertainty-Aware Offline Reinforcement Learning for Safe Medical Decisions." *Proceedings of the 5th Machine Learning for Health Symposium (ML4H), 2025*

Sophia Sun*, Aysin Tumay*, Sonia Fereidooni, Aaron Dumas, Rose Yu. "Designing Digital Twin to Support Safe Medical Decision-Making." *NeurIPS Workshop on Learning from Time Series for Health, 2025*

Sander Tonkens*, **Sophia Sun***, Rose Yu, Sylvia Herbert. "Scalable Safe Long-Horizon Planning in Dynamic Environments Leveraging Conformal Prediction and Temporal Correlations.*" IEEE International Conference on Robotics and Automation workshop on Long-term Human Motion Prediction (ICRA)*, 2023

Creative AI and Digital Humanities

Cheng, Mingyong, **Sophia Sun**, Han Zhang, and Yuemeng Gu. "Learning to Move, Learning to Play, Learning to Animate: a Multimedia Exploration of the More-than-human Intelligence.*" Proceedings of the ACM on Computer Graphics and Interactive Techniques 8, no. 3: 1-11 (SIGGRAPH 2025)* ***Award winner.***

Sofy Yuditskaya*, **Sophia Sun***, and Margaret Schedel*. "Synthetic Erudition Assist Lattice." *New Interfaces for Musical Expression (NIME), 2021*

Sofy Yudistkaya*, Derek Kwan*, **Sophia Sun***. "Karaoke of Dreams: A multi-modal neural-network generated music experience." *Proceedings of the Joint Conference on AI Music Creativity (CIMC+MUME),* 2020

Sophia Sun, Michael Scott Cuthbert. "Emotion painting: lyric, affect, and musical relationships in a large lead-sheet corpus." *Empirical Musicology Review 12.3-4*. 2018

Talks and Workshops

Learning to Move, Learning to Play, Learning to Animate San Diego, CA 2024
Contributed Talk at GenAI Summit 2025

Workshop on Statistical Frontiers in LLMs and Foundation Models Vancouver, BC 2024
NeurIPS Workshop Program Committee

Conformal Methods for Quantifying Uncertainty for Time-series Data Sunnyvale, CA 2023
Talk at AWS AI

Conformal Prediction for Spatio-Temporal Uncertainty Quantification Sunnyvale, CA 2023
Talk at UCSD CSE

Probabilistic Symmetry for Improved Trajectory Forecasting San Diego, CA 2022
Talk at UCSD CSE Research Open House AI spotlight

Reinforcement Learning 101 online 2020
Three-part workshop series for artists at Arts Counsel Korea (ARKO)

Natural Language Processing for Creatives Bombay Beach, CA 2020
Workshop at mars.college residency

Teaching

Lecturer & head TA, CSE251b Graduate level Deep Learning, UC San Diego — Spring 2025
Lecturer & head TA, CSE151b Deep Learning, UC San Diego — Spring 2023
Head TA, CSE 291 Deep Generative Models, UC San Diego — Fall 2022
Teaching Assistant, CSE101 Design & Analysis of Algorithms, UC San Diego — Summer 2022
Teaching Assistant, CSE20 Discrete Mathematics, UC San Diego — Summer 2022
Teaching Assistant, CSE251C Machine Learning Theory, UC San Diego — Spring 2021
Teaching Assistant, CSE291 Topics in Search and Optimization, UC San Diego — Winter 2020

Service and Leadership

Area Chair for NeurIPS 2026

Reviewer for ICML 2022 - 2025, NeurIPS 2022 - 2025, ICLR 2023 - 2025, JMLR 2025

Women in Data Science (WiDS), UC San Diego — Organizer, 2023 – 2024
UCSD ACM-W Student Chapter, UC San Diego — Chair, 2020 – 2022
Graduate Women in Computing (GradWIC), UC San Diego — Vice President, 2019 – 2022

Skills

Machine Learning: Algorithms for probabilistic spatial-temporal forecasting, anomaly detection, generative models, Transformers and LLMs, LLM Post-training and evaluation, Agentic systems

Uncertainty Quantification: UQ for time-series forecasting models and generative models, Bayesian methods and conformal prediction

Reinforcement Learning: Designing RL pipeline for real-world applications, Safe offline RL, multi-agent RL.

ML frameworks: **TensorFlow**, **Pytorch**, and **JAX**.

Data Science: AWS, **SQL, PostgreSQL**; distributed systems **Kubernectl**, **Hadoop on Spark**

Software Engineering: Competent in **Java** and **Python**, experience in **C, C++, R, Go, JavaScript/JQuery**